NEET
SAMPLE PAPER 1 - 2020

BIOLOGY
QUESTION PAPER

TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

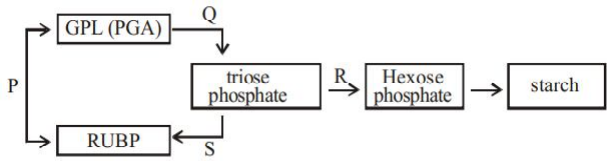
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1. Protoxylem is towards periphery and metaxylem towards pith. Xylem is said to be :
(A) Endarch (B) Polyarch (C) Misarch (D) Exarch

2. Heterosporous pteridophyte are :-
(A) Lycopodium and pteris (B) Selaginella and psilotum
(C) Selaginella and salvinia (D) Dryopteris and Adiantum

3. Non membranous organelle is :-
(A) Chloroplast (B) Nucleolus (C) Centriole (D) Both (B) and (C)

4. The diagram represent the calvin cycle :-



at which stage CO_2 incorporated ?
(A) P (B) Q (C) R (D) S

5. In which the following important intrinsic mechanisms provide auto regulation of glomerular filtration :-
(A) Myogenic mechanism (B) Cardiac mechanism (C) Tubular secretion (D) Kreb Henseleit cycle

6. Post pollination development of pollen grain is known as :-
(A) In-situ (B) Ex-situ (C) In-vitro (D) Precocious

7. All the following are uses of amniocentesis but one is misuse. Which one is misuse ?
(A) It is done to examine the sex of the foetus leading to increasing female foeticides.
(B) It is used to study metabolic defects of foetus like PKU (phenyl ketonuria)
(C) It is used to defect foetal abnormalities.
(D) It is used to detect genetic abnormalities.

8. Which of the following is best vector for plant and animal cells respectively :-
(A) Bacillus thuringiensis, Rosie (B) Pseudomonas putida, Microinjection
(C) Gene gun, Agrobacterium tumifacians (D) Agrobacterium tumifacians, Retroviruses

9. In his experiment S.L Miller observed the formation of :-
(A) Proteins (B) Simple Amino Acids (C) Lipids (D) Pigments, N_2 Bases

10. If biotic potential of a species is high than :
(A) Number of species in area is increase
(B) Species diversity increase in area
(C) Number of individuals of particular species are increase
(D) None of these

11. Ovary is one chambered but it becomes two chambered due to formation of false septum in:
(A) Brinjal and lemon (B) Mustard and capsella (C) Dianthus and primrose (D) Sunflower and marigold

12. Read the following statement. Find out how many statements are correct :-

(a) The leaves in Gymnosperms are well adapted to with stand extremes of temperature, humidity and wind.

(b) Gymnosperms are homosporous

(c) Unlike Bryophyta and pteridophyta in gymnosperms the male and female gametophyte do not have an independent free-living existence.

(d) In cycas male cone and megasporophyll are born on different trees.

(A) Three (B) Four (C) One (D) Two

13. Which of the following statement regarding chromosome is not correct ?

(A) At the time of cell division the chromatin material get condensed to form chromosome.

(B) At metaphase stage each chromosome is consist of two cylindrical structure is called chromatids.

(C) Kinetochore constitute actual site of attachment of spindle fiber to chromosome during cell division

(D) Telomere promote fusion of one chromosome to other chromosome

14. If pyruvic acid accumulated in cell then which regulation required :-

(A) Positive regulation By phasphofructokinase

(B) Alosteric modulation by pyruvic kinase enzyme

(C) Triose phosphate dehydrogenase enzyme inhibition.

(D) Link reaction acceleration

15. Identify the correct and wrong statements :-

(a) Human skeleton has 206 bones

(b) There are 12 pairs of ribs in human

(c) Sternum is present on the dorsal side of the body.

(d) All mammals (except of few) have 8 cervical vertebrae.

(e) Ribs are bicephalic

	correct	wrong
(A)	a, b, c	d, e
(B)	a, b, d	c, e
(C)	a, b, e	c, d
(D)	a, c, d	b, e

16. Polar nuclei are situated in the central cell :-

(A) Below the egg apparatus (B) Above the egg apparatus

(C) Below the antipodals (D) Near to micropyle

17. Which of the following is incorrectly matched ?

(A) Non-medicated IUD's – Lippes loop (B) Copper releasing IUD – LNG - 20

(C) Hormone releasing IUD – Progestasert (D) Saheli – Once a weak contraceptive pill

18. Recombinant DNA technology involves several steps in specific sequence. What is the correct sequence for r-DNA technology ?

a $\rightarrow$ Down stream processing. b $\rightarrow$ Cutting of DNA at specific location.

c $\rightarrow$ Obtaining the foreign gene product. d $\rightarrow$ Isolation of the genetic material.

e $\rightarrow$ Insertion of r-DNA into the host cell.

(A) b $\rightarrow$ d $\rightarrow$ e $\rightarrow$ a $\rightarrow$ c (B) d $\rightarrow$ b $\rightarrow$ e $\rightarrow$ c $\rightarrow$ a (C) d $\rightarrow$ e $\rightarrow$ d $\rightarrow$ a $\rightarrow$ c (D) e $\rightarrow$ d $\rightarrow$ a $\rightarrow$ c $\rightarrow$ b

19. Select the incorrect statement :-

(A) Lichen can be used as industrial pollution indicators.

(B) Evolution is a stochastic process based on chance event in nature and chance mutation in the organism.

(C) Evolution is a directed process in the sense of determinism.

(D) Excess use of herbicides, pesticides, etc has only resulted in selection of resistant varieties in a much lesser time scale.

20. How many statements are correct :

(a) Predators also help in maintaining species diversity in a community

(b) Decomposition is largely an oxygen requiring process

(c) Number of trophic levels in GFC is never restricted

(d) Some of organisms of DFC are prey to GFC animals

(A) Three (B) One (C) Two (D) All of the above

21. Which one of the following is enucleated dead cell ?

(A) Companion cell (B) Sieve tube (C) Guard cell (D) Vessel

22. Who discovered viroids :-

(A) Stanley B prusiner (B) T.O. Diener (C) D.J. Ivanowsky (D) W.M. Stanley

23. Which one is not found in matrix of mitochondria?

(A) 70-S Ribosome (B) Kreb's cycle enzyme

(C) Single strand DNA molecule (D) RNA molecule

24. ATP is injected in cyanide poisoning because it is :-

(A) Neccessary for cellular function (B) Neccessary for $Na^+ - K^+$ pump

(C) $Na^+ - K^+$ pump operates at the cell membraneous (D) ATP breakdown cyanide

25. During forensic investigation, a pelvic girdle was found, is of female. How does they concluded that it was of female not male ?

(A) Pelvic girdle diameter is less in female's than male's

(B) Pelvic diameter is more in female's than male's

(C) Female pelvis has one obturator foramen

(D) Pubic symphysis is not fused in females

26. Which of the following is formed as a result of triple fusion ?

(A) Fruit (B) Seed (C) Perisperm (D) Endosperm

27. Which one of the following is incorrect for garden pea :-

	Character	Genotype	Phenotype
(A)	Flower position	Homozygous or heterozygous	Terminal
(B)	Seed colour	Homozygous or heterozygous	Yellow
(C)	Pod colour	Homozygous or heterozygous	Green
(D)	Pod shape	Only Homozygous	Constricted

28. What out come is expected in offsprings if father is colourblind and normal mother's father was colourblind ?

(A) 50% daughter colourblind (B) All daughters colourblind

(C) All sons are colourblind (D) All sons are normal

29. Neopilina is a connecting link between :-
(A) Annelida and mollusca
(B) Annelida and Arthopoda
(C) Amphibia and reptilia
(D) Arthopoda and Mollusca

30. Fish die in polluted water by sewage due to :
(A) Pathogen (B) Biomagnification (C) Foul smell (D) Reduction in O_2

31. Select the mismatch from the following :
(A) Cruciferae – Tetradynamous
(B) Solanaceae – Perianth
(C) Liliaceae – Epiphyllous
(D) Fabaceae – Vexillary

32. The correct classification of Trygon is :-
(A) Chordata → Protochordata → Vertebrata → Pisces → Chondrichthyes
(B) Chordata → Ganathostomata → Pisces → Osteichthyes
(C) Choradata → Ganathostomata → Pisces → Chondrichthyes
(D) Choradata → Agnatha → Cyclostomata

33. Match the column-I and column-II and choose the correct option :-

	Column-I	Column-II
(a)	Synapsis	(i) Anaphase-II
(b)	Synthesis of RNA and protein	(ii) Zygotene
(c)	Action of enzyme recombinase	(iii) G_2-Phase
(d)	Centromere do not seprate but chromatids move towards opposite pole.	(iv) Anaphase-I
		(v) Pachytene

(A) a-iii, b-i, c-v, d-ii (B) a-ii, b-iii, c-v, d-iv (C) a-ii, b-i, c-iv, d-v (D) a-v, b-iii, c-iv, d-i

34. Plant hormone associated with climacteric respiration in :-
(A) Auxin (B) Cytokinin (C) Ethylene (D) Gibberellin

35. Typical cervical vertebra has :-
(A) Spinal foramen
(B) Transverse process
(C) Foramen Tranversarium
(D) Lamina

36. If the embryosac cell of an angiosperm has 12 chromosome then in the microspore mother cell will be :-
(A) 12 (B) 24 (C) 36 (D) 48

37. The phenotype is affected when :-
(A) Modified allele produces normal enzyme
(B) Modified allele produces non functional enzyme
(C) Unmodified allele produces normal enzyme
(D) Both 2 and 3

38. 'Restriction' in restriction enzyme refers to :-
(A) Cleaving of phosphodiester bond in DNA by enzyme
(B) Cutting of DNA at specific position only
(C) Prevention of multiplication of bacteriophages in bacteria
(D) All of the above

39. Antibodies which can pass through placenta, found abundantly in colostrum and involved in allergy respectively are

(A) Ig E, Ig A and Ig G (B) Ig G, Ig E and Ig A (C) Ig A, Ig E and Ig G (D) Ig G, Ig A and Ig E

40. Read statement (a-d) regarding population interaction :

(a) Human liver fluke has two intermediate host for survival

(b) Cuscuta is a parasitic plant lost its chlorophyll during evolution

(c) Female mosquito is considered as parasite

(d) Morphological and anatomical features are highly complex of endoparasites

Which of following statement is incorrect :

(A) a, b and c (B) a, c and d (C) b, c and d (D) c and d only

41. Which is not a stem modification :

(A) Tuber of potato (B) Offset of pistia

(C) Leaf bladder of utricularia (D) Corm of colocasia

42. Which one of the following phyla is correctly matched with its two general characteristics ?

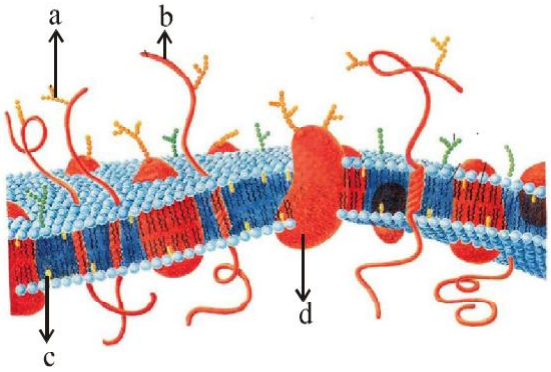
(A) Mollusca - Normally oviparous and development through a trochophore or veliger larva

(B) Arthropoda- Body divided into head thorax and abdomen and respiration by tracheae

(C) Chordata - Notochord at some stage and separate anal and urinary opening to the outside

(D) Echinodermata - Pentamerous radial symmetry and mostly internal fertilizatio

43. Recognise the figure and find out the correct matching:-



(A) a-Protein, b-Integral protein, c-Sugar, d-Cholesterol

(B) a-Sugar, b-Protein, c-Cholesterol, d-Integral protein

(C) a-Cholesterol, b-Sugar, c-Protein, d-Integral protein

(D) a-Integral protein, b-Sugar, c-Cholesterol, d-Protein

44. An enzyme that can stimulate germination of barley seeds is :-

(A) Lipase (B) Protease (C) Invertage (D) α-amylase

45. Cerebral hemispheres are connected by a tract of nerve fibres called as :-

(A) Corpus albicans (B) Corpus callosum (C) Thalamus (D) 2 and 3 both

46. When pollen grain of a flower pollinate the stigma of another flower of the same plant, it is called:-

(A) Xenogame (B) Herkogamy (C) Geitonogamy (D) Dichogamy

47. A cross is being given here :-

Red flowers White flowers — P

Crossing

Pink flowers — F_1

Selfing

A B C

If B stands for pink flower then

(a) What is the name of plant ? (b) What % of F_2 progeny will produce A ?

(A) (a) Garden pea (b) 25%

(B) (a) Snapdragon (b) 50%

(C) (a) Snapdragon (b) 25%

(D) (a) Garden pea (b) 50%

48. Name of microbe used for production of Swiss cheese :-

(A) Sacchromyces cerevisiae

(B) Propionibacterium sharmanii

(C) Monascus purpureus

(D) Lactobacillus

49. Metastasis can be defined as

(A) the spread of cancer from one person to another

(B) the spreading process of sexually transmitted diseases.

(C) the spreading process of communicable diseases from one organ to another.

(D) the spreading process of cancer from one organ to another through body fluids.

50. Slash and burn agriculture commonly called as jhum cultivation in North-Eastern state of India related to :

(A) Deforestation (B) Afforestation (C) Crop rotation (D) Dry farming

51. Term "Taxonomy" was given by :-

(A) Carolus Linnaeus (B) Gaspard Bauhin (C) A.P. Candolle (D) J. Huxley

52. Cylindrical bodies, worm like marine animals belong to the phylum :-

(A) Echinodermata (B) Mollusca (C) Coelenterata (D) Hemichordata

53. Which of the following is a rapid type of absorption :-

(A) Passive absorption (B) Active absorption (C) Salt absorption (D) Root absorption

54. A doctor advises a patient to include yellow fruits carrots and butter in his diet. What deficiency disease do you think the patient is suffering from :-

(A) Colour blindness (B) Scurvy disease (C) Night blindness (D) Marasmus disease

55. Forebrain does not include :-

(A) Cerebrum (B) Thalamus (C) Hypothalamus (D) Corpora quadrigemina

56. A aquatic plant, water lily, is pollinated by :-

(A) water (B) air (C) insect (D) birds

57. Which one of the following is not a mendelian disorder?

(A) Haemophilia (B) Sickle cell anaemia (C) Phenylketonuria (D) Down's syndrome

58. In cells of superfemale with 47 chromosomes (44 + XXX), visible barr bodies are :-
(A) 1 (B) 0 (C) 2 (D) 3

59. Mark the incorrect combination
(A) Aquaculture- crab, lobsters and edible oyster
(B) Fresh water fishes- Catla, rohu, comman carp and pomfrets
(C) Marine fishes- Hilsa, Sardine, Mackerel
(D) MOET- Cattle, sheep rabbit, buffaloes and mares

60. Logistic growth and exponential growth formula is :
(A) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN\left(\frac{N-K}{N}\right)$ (B) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$; $\frac{dN}{dt} = rN$
(C) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN$ (D) $\frac{dN}{dt} = rN$; $\frac{dN}{dt} = rN\frac{(K-N)}{N}$

61. In Magnifera indica, indica represents :-
(A) Generic name (B) Name of outhor (C) Specific epithet (D) Subspecific name

62. Match the columns I, II, and III and choose the correct combination from the options given :-

	Column-I	Column-II	Column-III
A.	Testes	(i) 2nd-6th Segments	(P) 1 Pair
B.	Ovaries	(ii) 4th- 6th Segments	(Q) 1 Pair
C.	Spermathecae	(iii) 6th- 7th Segments	(R) Double
D.	Mush room gland	(iv) 6th- Segment	(S) Single

(A) A-(ii)-R, B-(i)-Q, C-(iii)-P D-(iv)-P (B) A-(ii)-P, B-(i)-Q, C-(iv)-P, D-(iii)-S
(C) A-(i)-Q, B-(ii)-P, C-(iv)-R, D-(iii)-S (D) A-(iii)-P, B-(iv)-R, C-(i)-Q, D-(ii)-S

63. Cell A and B are adjacent plant cells. In cell A,
$\psi_s = -20$ bars and $\psi_p = 8$ bars. In cell B, $\psi_s = -12$ bars and $\psi_p = 2$ bars. Then :-
(A) Water moves from cell A to cell B
(B) There is no movement of water between cells A and cell B.
(C) Water moves from cell B to cell A
(D) Equal amount of water is simultaneously exchanged between cell A and cell B.

64. Which one of these statement about intrinsic factor is not true ?
(A) It is a glycoprotein (B) It is secreted by gastric gland
(C) It helps to prevent pernicious anaemia (D) It promotes absorption of vitamin B12in the intestine

65. Disease caused by hypo and hypersecretion of collip's hormone is :-
(A) Tetany and Osteoporosis (B) Osteoporosis and Tetany
(C) Tetany and Asphyxia (D) Asphyxia and Cretinism

66. The Graafian follicle ruptures to release _____ from the ovary by the process called ovulation.
(A) Primary oocyte
(B) Secondary oocyte after completing meiosis-II.
(C) Secondary oocyte after completing meiosis-I & with the release of Ist polar body.
(D) Mature ovum

67. Select incorrect statement regarding DNA replication ?
(A) Leading strand is formed in 5' → 3'direction.
(B) Okazaki fragments are formed in 5' → 3' direction
(C) DNA polymerase catalyzes polymerisation in 5' → 3' direction
(D) Continuous strand synthesizes from 5' → 3' and discontinuous strand from 3' → 5' direction

68. A gene showing codominance has :
(A) both alleles independently expressed in the heterozygote
(B) one allele dominant on the other
(C) alleles tightly linked on the same chromosome
(D) alleles that are recessive to each other

69. Which of the following species of plasmodium cause most serious and fatal malaria.
(A) P.vivax (B) p. falciparum (C) P. malariae (D) P. ovale

70. In equation $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$; here K is defined by :
(A) Maximum population density achievable under available resources
(B) Carrying capacity of habitat
(C) Both (A) and (B)
(D) Maximum interinsic rate of growth

71. Which of the following statement is incorrect ?
(A) Fungi prefer to grow in cold and humid places.
(B) Karyogamy in fungi occurs after plasmogamy
(C) Yeast are used to make bread and beer.
(D) Fungi bodies consist of long slender thread like structure called mycelium

72. Virus particle outside the host body is called
(A) Virion (B) Prion (C) Viriods (D) Bacteriophage

73. Select the correct match :-

A	Nitrosomonas	Nitrite to nitrate
B	Thiobacellus	Denitrification
C	Nastoc Free-living	nitrogen-fixer
D	Azotobacter	Anaerobic nitrogen-fixer

(A) A and B (B) C and D (C) B and C (D) B and D

74. Which type of white blood cells are concerned with the phagocytic in nature and maximum in number ?
(A) Monocytes (B) Eosinophils (C) Basophils (D) Neutrophils

75. Match the correct sets :-

	Endocrine	Hormone	Deficiency
(a)	Adenohypophysis	Vasopressin	Diabetes Mellitus
(b)	Parathroid gland	Parathormone	Tetany
(c)	Pancreas	Insulin	Diabetes Mellitus
(d)	Thyroid gland	Calcitonin	Acromegaly

(A) a, d (B) b, d (C) b, c (D) a, b

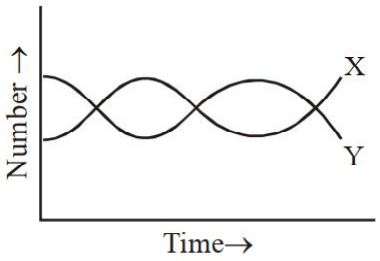
76. Signals from fully developed foetus and placenta ultimately leads to parturition which requires the release of :
(A) Estrogen from placenta (B) Oxytocin from foetal pituitary
(C) Oxytocin from maternal pituitary (D) Relaxin from leydigs cells

77. If the base sequence of a codon in mRNA is 5'-AUG-3', the sequence of tRNA pairing with it must be :-
(A) 5'-AUG-3' (B) 5'-UAC-3' (C) 5'-CAU-3' (D) 5'-GAU-3'

78. C-DNA molecule synthesized against m-RNA using reverse transcriptase has a length which is :-
(A) Lesser than in vivo gene (B) Lesser than mRNA
(C) Same as in vivo gene (D) Greater Than mRNA

79.. A person is having dry, scaly lesions on various parts of the body such as skin, nails and Scalp. These lesions are accompanied by intense itching. This disease may be caused by
(A) Microsporum and Entamoeba (B) Trichophyton and microsporum
(C) Epidermophyton and Plasmodium (D) Entamoeba and Plasmodium

80. What type of interaction shows in this graph between two organism X and Y :



(A) Mutualism (B) Commensalism (C) Parasitism (D) Predation

81. Read the following characters find out which one is not related with Gymnosperm :-
(A) Association with N_2 fixing cyanobacteria (B) Mycorrahiza
(C) More than one archegonia (D) Independent free living gametophyte

82. Read the following statement carefully (A-D) :-
(a) Anything less than a complete structure of a cell does not ensure independent living.
(b) Ribosome are membrane bound organelles found in prokaryotic and Eukaryotic cell.
(c) Cytoplasm is the main arena (Zone) of cellular activities in both plant and animals cell.
(d) Movement of water by diffusion is called osmosis.
How many statement are correct ?
(A) One (B) Two (C) Three (D) Four

83. The rate of photosynthesis does not depend upon :-
(A) Light duration (B) Light intensity (C) Light quality (colour) (D) Temperature

84. Which blood vessels contains the more amount of urea ?
(A) Hepatic vein (B) Renal vein (C) Hepatic Artery (D) Renal Artery

85. Hormone that initiates ejection of milk and promote production of antibodies to provide humoral immunity :-
(A) Oxytocin and Thymosins (B) Prolactin and Thymosins
(C) Oxytocin and Cortisol (D) Follicle stimulating hormone and Cortisol

86. Identify the hormones that are secreted is large amount prior to ovulation.

(i). LH (ii). FSH

(iii). Progesterone (iv). Estrogen

(A) Only (i) (B) Only (i) and (iv) (C) (i), (ii) & (iv) (D) (i), (ii), (iii), (iv)

87.

A	B
Occur in pairs	Occur in pairs
Segregate at the time of gamete formation such that only one of each pair is transmitted to a gamete	Segregate at gamete formation and only one of each pair is transmitted
Independent pairs segregate independently of each other	One pair segregates independently of another pair

Select out the incorrect conclusion :-

(A) All the statements of column A are true for chromosome

(B) All the statements of column B are always true for chromosome

(C) All these statements of column A are always true for Gene

(D) All the statements of column B are always true for Gene

88. Which of the following statement is false :-

(A) The 1st cellular form of life did not possibly originate till About 2000 mya.

(B) The geological history of earth closely correlates with biological history of earth.

(C) According special creation theory earth is about 4.5 billion year old.

(D) The universe is very old-almost 20 billion years old.

89. How many statements are correct :

(a) Domestic sewage contains biodegradable organic matter.

(b) A scruber remove gases like sulphur dioxide

(c) Electrostatic precipitates can remove 99% particulate matter present in exhaust from thermal power plant.

(d) Particulate size 2.5 μm or less in diameter are not harmful for human health

(A) Two (B) One (C) Four (D) Three

90. Desert, Grassland, Tropical forest, Temperate forest, Coniferous forest and Tundra. Which of the following biome research area of Ramdeo Mishra ?

(A) Tundra and Temperate forest (B) Desert and Grassland

(C) Tropical forest and Desert (D) Tropical forest and Grassland

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